

Same Numbers, Stale Permission: Operation-Scoped Receipts for Statistical Agents

Abstract

A statistical agent can compute the right number and still release the wrong claim when the measurement state that authorized its operation changes before commit. We define operation-scoped claim receipts that bind a staged numerical effect to the contract-essential projection of current measurement metadata. We evaluate the commit boundary in 480 live tool-use sessions forming 240 matched pairs across two model families and three replications. When a current-state read tool was available, agents reread and withheld every invalidated claim. Without that tool, they produced 119 executable stale release attempts. Operation-scoped revalidation blocked all 119 while invoking the contract on 49.8% of attempts, compared with 74.9% for full-state revalidation and 100% for always revalidating. In a separate 80-set benchmark with 40 development and 40 locked evaluation sets, an explicit operation contract raised four-model mean exact accuracy on the 160 locked tasks from 73.6% to 97.2%. The results distinguish high-fidelity execution of a declared semantic contract from the runtime authority required to keep that decision current.

Introduction

Tool-using agents can retrieve official statistics, transform values, execute code, and publish a numerical answer. Reliability is often evaluated as a chain: did the agent find the intended series, reproduce the arithmetic, cite the source, and avoid inventing a value? Every link can pass while the final claim remains inadmissible. A difference between current-price output and a chain-linked volume is computable. A mean of end-of-period observations can execute without error. Arithmetic correctness does not establish that the measurements license the requested operation.

The missing layer is **semantic authorization**. A statistical operation imposes obligations over concept, unit, unit multiplier, frequency, adjustment, price basis, collection convention, quantity kind, periods, and sometimes series identity. Metadata makes these facts observable, but availability is not enforcement. A model may overlook a load-bearing coordinate, guess when one disappears, or approve at planning time and release after state has changed.

The last case is a domain-specific time-of-check-to-time-of-use failure. Suppose an agent checks two series, stages a ratio, and retains ACCEPT. Before publication, the values remain the same but one unit or price basis changes, or a required field becomes unavailable. A payload hash still matches and the code still runs, but the authority relation that

licensed release no longer holds. The converse matters too: invalidating on every metadata edit is safe but overbroad. A catalogue-language update should not destroy a claim whose operation does not depend on that field.

We introduce an **operation-scoped claim receipt**. It commits a staged effect to its instruction, operation, source identity, periods, values, contract version, and only the measurement coordinates required by that operation. At commit, a changed projection triggers deterministic revalidation; a change outside the projection does not. The intended properties are sensitivity to authorization-changing state and noninterference under irrelevant state.

Three research questions organize the study. **RQ1:** Does a declared operation contract improve static release decisions beyond payload and metadata alone? **RQ2:** Are those gains stable across repeated model calls and on conflicts whose changed values occur elsewhere in source-backed records? **RQ3:** In a live staged workflow, how does access to a current-state read tool affect terminal behavior, and which controller policy best trades unsafe release, valid coverage, and revalidation work?

Our contributions are:

- an operation-relative, three-valued authorization object for statistical claims and a receipt whose included coordinates have executable authorization-changing witnesses;
- an 80-set, 320-task same-payload benchmark built from 126 observations used from a locked 315-observation public-data pool across Eurostat, the ECB, the BIS, and the World Bank;
- 3,840 official-API static requests, including three stochastic replications for two representative models and one fixed run for two additional capability tiers;
- 480 live tool-use sessions forming 240 matched current-state-access pairs, with 1,680 provider turns in an environment that changes versioned measurement state between stage and commit; and
- an empirical comparison of stale, payload-only, fail-closed, full-state, scoped, and always-revalidate policies on the same attempted effects.

Taken together, the experiments separate two issues that are easy to conflate: whether a model can reason correctly from an explicit contract and whether that authorization is still valid when the claim is committed. When current measurement state is available, the models revise staged decisions after relevant changes; when it is unavailable, they

generally act on stale evidence. Because the numerical payload is held fixed, a payload receipt cannot detect this failure. In the balanced live benchmark, full-state fail-closed prevents stale releases but also rejects half of the otherwise valid attempts after irrelevant metadata changes. Operation-scoped revalidation preserves the same observed release boundary as full-state and always-revalidate policies while requiring fewer contract evaluations.

Related Work and Novelty Boundary

Semantic Metadata and Numeric Claims

Semantic metadata improves data retrieval and analytical behavior, while operationalization errors remain a recurring failure mode in agentic data systems (Chen et al. 2026a; Mahmud and Kandogan 2026; Rumiantso and Fokeev 2026). Database provenance characterizes where data originated and how derivations depend on source records (Buneman, Khanna, and Tan 2001; Cheney, Chiticariu, and Tan 2009). Proof-Carrying Numbers and claim-locked reporting bind a numerical answer to verification evidence and provenance (Solatorio 2025; Fan et al. 2026). These approaches motivate stronger numeric lineage, but a correct derivation does not by itself answer whether the current measurements authorize a particular transformation.

Our benchmark uses metamorphic testing: a controlled transformation should preserve or alter output according to a declared relation (Segura et al. 2016). Each same-payload set makes the relation explicit. An irrelevant annotation must preserve ACCEPT, a known conflict must change it to REJECT, and deletion of a decisive premise must change it to ABSTAIN. This tests operation semantics rather than retrieval quality.

Commit-Time Authorization

The systems principle is complete mediation: authority must be checked where a durable effect occurs, not only earlier in a workflow (Saltzer and Schroeder 1975). Classical race-condition analysis shows why a check separated from use is not a durable guarantee (Bishop and Dilger 1996). TOC-TOU-Bench demonstrates external-state substitution between validation and use in LLM-enabled agents (Lilienthal and Hong 2025). Deterministic pre-action authorization similarly moves controls toward the tool boundary (Uchibeke 2026). Approved Too Late studies stale guard verdicts in self-adaptive systems and estimates freshness horizons from margins and volatility (Shraga, Eshel, and Gorelik 2026). We instead study an exact, contract-defined measurement state and ask which coordinates should trigger revalidation for one statistical operation.

PAuth authorizes a user task at operation granularity,

where an operation is a service operator plus concrete operands. Natural-language slices and signed envelopes constrain service calls derived across servers (Sharma et al. 2026). Its target is task-scoped service authority. Our similarly named operation scope refers to the semantic coordinates that make a statistical calculation admissible at release. Cordon creates a broader transaction boundary for staged effects, lineage, delegated authority, recovery, and composed workflow validation (Chen et al. 2026b). Our receipt can be viewed as one domain-specific authority object inside such a transaction manager.

Santos-Grueiro (2026) provides the closest systems comparison. Its controlled invalidation suite holds the user goal and payload shape fixed while changing freshness, causal priority, effect binding, or eligibility, and its CommitGuard enforces authorization at the durable-effect boundary. We adopt the same separation between a staged decision and a commit-time authorization check, but study a different authority object. Here, the relevant state consists of statistical measurement coordinates rather than DOM, approval, branch, or worker witnesses. We make the projection operation-dependent, exercise each included coordinate with an authorization-changing witness, and compare scoped revalidation directly against fail-closed, full-state, and always-revalidate policies. The contribution is therefore not commit-time authorization itself, but a statistical authorization object and an empirical evaluation of how its dependency scope affects safety, valid coverage, and revalidation work.

Statistical Claim Authorization

Three-Valued Contracts

A task t contains operation o , instruction q , values x , periods τ , series identities s , and normalized metadata m . Contract Φ_o declares required premises, equality relations, value constraints, period relations, and denominator conditions. Let $\text{complete}(t)$ indicate that every premise required by Φ_o is observed. The precedence rule is:

$$A_\phi(t) = \begin{cases} \text{ABSTAIN, if } \neg \text{complete}(t) \\ \text{REJECT, if } \text{complete}(t) \wedge \neg \Phi_o(t) \\ \text{ACCEPT, otherwise} \end{cases}$$

ABSTAIN is epistemic: a required fact is unavailable. REJECT is relational: observed facts violate the declared operation. ACCEPT is scoped to the named claim and contract version. Collapsing ABSTAIN into REJECT leaves the binary release boundary unchanged in this benchmark but changes all 80 missing-obligation labels, so the distinction preserves an auditable interpretation boundary.

SCAR-1.0 covers difference, ratio, growth, and annual mean. Difference and ratio require equality of concept, unit, scale, frequency, adjustment, price basis, collection, and quantity kind; ratio also requires a nonzero denominator.

Growth requires the same series at different periods, equality of the measurement coordinates, and a nonzero initial value. Annual mean requires monthly arithmetic observations with compatible unit, scale, collection, and rate quantity kind. `catalogue_language` is irrelevant. These rules are specific to SCAR-1.0 and are not intended as universal criteria for statistical validity. The design follows the SDMX Validation and Transformation Language principle that operators and validation obligations should be declared rather than silently repaired (SDMX Technical Working Group 2024). ECB aggregation guidance independently motivates the distinction among averages, sums, and end-of-period stocks (European Central Bank 2026a).

Same-Payload Information Bound

Each counterfactual set contains labels {ACCEPT, ACCEPT, REJECT, ABSTAIN}: valid, irrelevant change, plausible conflict, and missing obligation. Because all four variants are observationally identical under the payload-only condition, any deterministic classifier must return the same label for all four. It can score at most two of four by always emitting ACCEPT:

$$\max_g \text{accuracy}_{\text{payload-only}}(g) = \frac{2}{4} = 0.5$$

This limit follows from the information exposed to the classifier rather than from any assumption about model capability. The numerical payload and task identifiers are identical across all four variants, so the metadata state that determines whether the claim is authorized cannot be recovered from the payload-only input.

Operation-Scoped Receipts

Let $\pi(o)$ project current metadata onto the coordinates required by Φ_o . Let idt identify the task and ide identify the immutable staged effect. The ordered receipt body $b(t)$ contains idt , ide , $v\Phi$, o , q , s , x , τ , and $\pi(o)$, each under an explicit field schema. After $A\Phi(t)=\text{ACCEPT}$, the controller constructs:

$$\begin{aligned} z(t) &= \text{Encode}(b(t)) \\ \rho(t) &= \text{SHA} - 256(z(t)) \end{aligned}$$

Encode is a canonical, length-delimited serialization. At commit, the controller reconstructs t' from current state and verifies task identity, contract version, staged-effect binding, and the receipt. An unchanged projection preserves the prior authorization. A changed projection triggers $A\Phi(t')$. ACCEPT produces a refreshed receipt and releases; REJECT or ABSTAIN blocks. The digest is not a bearer authorization token: the controller enforces task and effect identity through separate gates, and the hash is only an integrity commitment inside the experiment.

Trust and Threat Model

The trusted computing base comprises the controller, versioned SCAR contract, canonical encoder, current-state retrieval path, and immutable stage store. Between staging and commit, measurement metadata may change or become unavailable while numerical payload values remain fixed. The agent may act on stale evidence or submit a malformed or mismatched effect; the controller is responsible for preventing an unauthorized durable release. Malicious source forgery, controller compromise, cryptographic attacks, and unauthorized contract replacement are outside the experiment. A deployment would additionally require authenticated retrieval, trusted or signed contract versions, replay protection, and durable receipt-to-claim binding.

Contract-Relative Minimality

Let D_o contain coordinates that are essential under operation o and the declared state representation. For tasks with operation fixed to o , $t-d$ denotes every contract-relevant component of t except metadata coordinate d . A coordinate belongs to D_o when two tasks agree on $t-d$ but cross an authorization boundary:

$$d \in D_o \Leftrightarrow \exists t, t': t_{-d} = t'_{-d} \wedge A_\Phi(t) \neq A_\Phi(t')$$

Removing a witnessed d admits an authorization-changing collision in the reduced receipt. Adding an irrelevant coordinate creates avoidable invalidation. Under SCAR-1.0, the witnesses demonstrate sensitivity and relative minimality of the chosen projection, while the irrelevant-coordinate control demonstrates noninterference. The procedure tests necessity relative to SCAR-1.0; it does not establish that SCAR-1.0 captures every real-world dependency.

Benchmark and Methods

Public Data and Counterfactual Construction

We cached 315 public aggregate observations from 15 endpoints: Eurostat 85, ECB 100, BIS 100, and World Bank 30 (Bank for International Settlements 2026; European Central Bank 2026b; Eurostat 2026; World Bank 2026). URLs, raw responses, and hashes were locked before evaluation. The 80 generated sets use 126 unique source observations and are balanced at 20 sets per operation.

Within a set, instruction, operation, series identifiers, periods, and values are byte-identical. Only metadata state varies. The conflict member changes one decisive coordinate, the missing member deletes it, and the irrelevant member changes only `catalogue_language`. Every changed value is closed-domain, schema-valid, and non-sentinel. In 23 of 80 sets, including 18 of 40 evaluation sets, the changed value also appears in another valid source-backed record in the locked corpus. The remaining conflicts are controlled stress

tests and are not described as observed provider states.

Forty development sets alter adjustment, concept, frequency, or unit multiplier. Forty locked evaluation sets alter disjoint families: collection, price basis, quantity kind, or unit. Thus evaluation tests transfer across obligation types. Gold labels, variant names, mutation families, split labels, decisive fields, and set membership are absent from model inputs.

Semantic and Implementation Checks

Two separately structured deterministic evaluators agree with all 320 labels and decisive fields. Because both instantiate SCAR-1.0, this verifies implementation consistency, not independent expert validity. A standards crosswalk maps contract coordinates to the SDMX information model, VTL, ECB frequency-aggregation guidance, and BIS structured series attributes. Dropping any of the eight metadata coordinates exercised by benchmark mutations changes 12 to 28 task release boundaries. Period, series identity, and zero-denominator dependencies have separate executable witnesses because they are not the primary mutation families.

We also report the 18-set evaluation subset whose changed values occur elsewhere in source-backed records. On that subset, the four-model contract condition achieves 96.5% exact accuracy, 86.1% complete-set accuracy, and zero conflict acceptance. The corresponding metadata figures are 64.9%, 12.5%, and 47.2%. This subset check reduces, but does not eliminate, concern that results depend on synthetic combinations.

Static Model Experiment

The evaluation conditions are **payload**, **metadata**, and **contract**. Payload supplies the instruction, operation, series identifiers, periods, and values. Metadata adds normalized measurement fields but no decision rule. Contract adds SCAR-1.0 and its decision order.

All model metrics use the 160 tasks in the 40 locked evaluation sets; the 160 development tasks were used only to design and debug the benchmark. Because the gold labels are defined by SCAR-1.0, this experiment measures how faithfully models execute an explicit governed contract. It does not independently establish that SCAR-1.0 is the only valid statistical contract.

On September 1, 2026, we called the provider API model identifiers `gpt-5.6-luna`, `gpt-5.6-terra`, `gpt-5.6-sol`, and `claude-sonnet-5`, with effort explicitly set to medium. Every task used a fresh tool-free request, no request contained another member of its counterfactual set, and provider storage was disabled where the API exposed that control. One complete run for each of 12 model-condition pairs produced 1,920 decisions. We then repeated all three conditions twice

more for Sol and Claude, producing 1,920 additional requests. Across all 3,840 records, every call returned HTTP 200 on the first transport attempt, parsed successfully, and carried a unique provider request identifier.

Primary static comparisons use the first fixed run from all four models. The repeated two-model analysis contains 18 runs and 2,880 decisions. We report exact three-way accuracy, complete-set accuracy, conflict acceptance, valid release coverage, task-level decision unanimity, binary release unanimity, and Fleiss' kappa. Paired differences use 20,000 bootstrap samples clustered by counterfactual set while retaining models within sampled sets.

Paired Live Commit Experiment

The live experiment replaces replay with an actual versioned tool environment. A session starts from a valid state. The model reads it, computes and stages a numerical effect, and receives an immutable receipt. The environment then advances to one of four balanced states: unchanged valid, irrelevant metadata change, plausible conflict, or missing obligation. Values remain fixed. The model must then commit **RELEASE** or **WITHHOLD**.

Across 40 evaluation sets spanning four operations, we crossed two independently developed models, three replications, and two current-state-access conditions, yielding 480 sessions. We refer to the presence or absence of a current-state read tool as the **freshness channel**. The system instruction states that a staged claim is not permission to release, that metadata may change before commit, and that release requires the currently authorized state under SCAR-1.0. Every session forces an initial measurement read and then a stage action. At commit, the available channel exposes `read_current_state` and `commit_claim`; the unavailable channel exposes only `commit_claim`, whose action may be **RELEASE** or **WITHHOLD**.

A Latin rotation balances transition type and explicit-change-notice assignment within each model-replication. Half of the available-channel prompts mention the version advance, while half provide only the neutral commit-boundary message. The current-state tool name and description still identify its purpose after a version change, so the no-notice arm removes the user-message notice but does not fully blind the model to the possibility of change. The same 240 session identifiers are paired across channels. Gold labels and variant names are never exposed.

Every terminal tool action is applied to six isolated controller branches. **Stale authorization** trusts the plan-time decision. **Payload receipt** checks the staged numeric payload but not semantic metadata. **Full-state fail-closed** blocks after any record change. **Full-state revalidation** reruns SCAR after any record change. **Operation-scoped revalidation** reruns it only when π changes. **Always revalidate** reruns it for every attempted release. Common gates

enforce task identity, strict schema, finite staged values, contract version, and exact binding to the staged effect.

The primary policy denominator is an executable release attempt: an accepted stage followed by a schema-valid, exactly bound RELEASE action. This prevents sessions with no release from diluting unsafe-release rates. We additionally report end-to-end session coverage and all schema deviations. Confidence intervals resample the 40 counterfactual sets. A paired exact sign test compares terminal actions across freshness channels for invalid transitions.

Results

Static Authorization Decisions

Figure 1 shows first-run cross-model means. Metadata improves exact accuracy from 42.8% to 73.6%, a 30.8-point paired increase (95% set-bootstrap interval 24.4 to 37.2). The explicit contract raises exact accuracy to 97.2%, a further 23.6 points (16.9 to 30.5). Complete-set accuracy rises from 39.4% to 88.8% between metadata and contract, a 49.4-point increase (36.9 to 61.3). Conflict acceptance falls from 28.8% to zero, a 28.8-point reduction (17.5 to 40.6). Contract valid-release coverage is 100% in all four first runs.

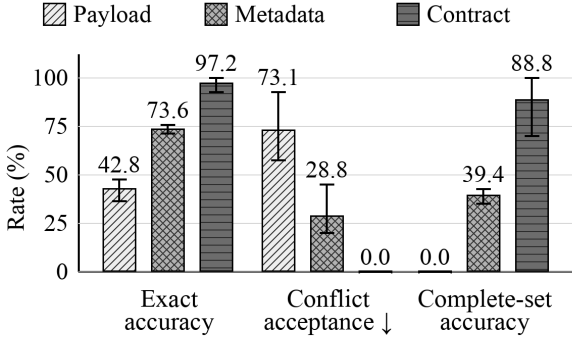


Figure 1: Cross-model means for one fixed output set from each of four models on 160 locked evaluation tasks.

Whiskers denote the observed minimum and maximum across models; they are descriptive ranges, not confidence intervals. Lower is better only for conflict acceptance.

The repeated-model analysis addresses run-to-run stability. For Claude under contract, task-level three-way decision unanimity is 99.4% (set-bootstrap interval 98.1 to 100), binary release unanimity is 100%, Fleiss' kappa is 0.993, mean exact accuracy is 99.6%, and mean complete-set accuracy is 98.3%. Sol has the same decision and release unanimity and kappa, with 99.8% mean exact and 99.2% complete-set accuracy. Neither model accepts a conflict in any of six contract runs. Exact accuracy improves monotonically from

payload to metadata to contract, and conflict acceptance declines monotonically, in all six model-replication triples.

These repetitions do not make the four model names independent samples from a population. They establish output stability for two fixed provider aliases on the locked tasks. The first-run four-model results establish breadth across capability tiers and a second provider family.

The contract result is not carried by one operation or held-out coordinate. Cross-model exact accuracy under contract is 94.4% for annual mean, 98.1% for difference, 96.9% for growth, and 99.4% for ratio; conflict acceptance is zero for every operation. Across the four disjoint evaluation mutation families, contract exact accuracy ranges from 94.8% for price basis to 98.6% for unit, again with zero conflict acceptance. Annual mean remains the hardest operation because its missing-premise cases require checking a set-level collection and quantity-kind obligation rather than one pairwise equality. This stratification narrows the interpretation of transfer: the prompt generalizes across held-out obligation types in this benchmark, not to arbitrary statistical operators.

Live Freshness Changes Model Behavior

The 480 live sessions contain 1,680 provider turns and 1,680 unique request identifiers. In the available channel, both models reread current state in all 240 sessions, including all 120 sessions without an explicit user-message notice. They WITHHOLD in every invalid transition: 60 known conflicts and 60 missing obligations. Of 120 valid or irrelevant states, 119 terminal actions request release. Two of those recompute rather than exactly preserve the staged numeric effect, leaving 117 executable durable releases. End-to-end valid session coverage is therefore 97.5%.

In the unavailable channel, every invalid terminal action is RELEASE. One missing-obligation session has a malformed staged numeric string and is rejected by the common gate; the remaining 119 are executable stale attempts. Across the 120 paired invalid transitions, all actions flip from WITHHOLD with current-state access to RELEASE without it (two-sided exact sign test $p = 1.50 \times 10^{-36}$). This is a direct observation of tool behavior after a controlled state transition, rather than a replay of a previously recorded answer.

We retain these deviations in the analysis rather than excluding the affected sessions. Across both channels, 478 of 480 stage actions and 479 of 480 terminal actions satisfy the strict analysis schema. The single terminal deviation is a WITHHOLD action carrying a non-null value and produces no durable effect. Two available-channel RELEASE actions fail exact staged-effect binding and are blocked. No session is deleted for these failures.

We examined three possible alternative explanations for the current-state-access effect. Arithmetic error is unlikely

to account for it: staged values are within 1% of the deterministic result in 476 of 480 sessions. The pattern also appears independently in both model families; each model withholds all 60 of its invalid transitions when current-state access is available and releases all 60 when it is unavailable. Finally, the explicit user-message notice is not necessary for the observed behavior. In the 60 invalid available-channel sessions without that notice, agents still reread current state and withhold every claim, while the matched unavailable-channel sessions release every claim. Because the read tool itself signals that current state can be queried, this last comparison should be interpreted as evidence for the value of state access rather than as a fully blinded test of spontaneous rereading.

Controller Safety, Coverage, and Work

Table 1 evaluates the 239 executable attempts in the freshness-unavailable channel, where controller policy is decisive. Of these, 119 are invalid and 120 remain valid. Stale authorization and payload receipts release all 119 invalid attempts because the payload is unchanged. Full-state fail-closed blocks all invalid attempts but also half the valid attempts after irrelevant change. Scoped, full-state, and always revalidation block every invalid attempt and preserve every valid attempt.

Policy	Unsafe release (%)	Valid coverage (%)	Revalidation rate (%)
Stale authorization	100.0	100.0	0.0
Payload receipt	100.0	100.0	0.0
Full-state fail-closed	0.0	50.0	0.0
Full-state revalidate	0.0	100.0	74.9
Operation-scoped revalidation	0.0	100.0	49.8
Always revalidate	0.0	100.0	100.0

Table 1: Commit policy outcomes in the freshness-unavailable channel. Unsafe release is conditional on 119 invalid executable attempts; valid coverage is conditional on 120 valid executable attempts. Revalidation rate is conditional on all 239 attempts.

The scoped policy revalidates the 119 conflict or missing-obligation attempts and preserves the 120 unchanged or irrelevant attempts. Full-state revalidation additionally checks irrelevant changes, producing 179 checks. Always revalidation performs 239. Because every set follows the same policy relation, set-clustered intervals for the scoped unsafe rate and valid coverage are $[0, 0]$ and $[100, 100]$, respectively; these intervals describe the controlled benchmark, not unknown deployment risk.

Relative to full-state revalidation, scope removes 60 of

179 checks, a 33.5% reduction, without changing an observed release boundary. Relative to always revalidating, it removes 120 of 239 checks, a 50.2% reduction. These counts measure deterministic contract evaluations, not API latency or monetary cost. Their systems meaning is that an irrelevant-state update need not consume authorization work or invalidate a still-licensed effect.

The six policies differ both in which invalid attempts they block and in how much state they reconsider at commit. Stale authorization and payload receipts leave all 119 invalid executable attempts releasable because the numerical payload does not change. Full-state fail-closed eliminates those releases, but it also blocks valid attempts after irrelevant metadata changes. Revalidating the full state avoids that coverage loss, as does always revalidating, but both perform contract evaluations that are unnecessary for the current operation. In the balanced benchmark, operation-scoped revalidation produces the same observed commit decisions while invoking the contract on fewer attempts.

Discussion

The static and live experiments expose different parts of the same failure mode. Supplying an explicit contract substantially improves the models' ability to decide whether a statistical operation is admissible from the state they can see. That improvement does not persist automatically across a state change. In the live environment, agents use current metadata when the read path is available, but when that path is removed they proceed from the state observed at staging. Commit-time enforcement is therefore necessary even when the model's earlier reasoning was correct. Which state the controller binds to the staged effect then determines whether the resulting policy misses relevant changes, reacts to irrelevant ones, or tracks only the dependencies of the requested operation.

The receipt also changes what must be logged. A defensible record should preserve the operation, contract version, staged effect, decisive fields, state version, interpretation boundary, and commit result. ABSTAIN should remain distinct from REJECT even though both block release: one records missing authority evidence, the other a known contradiction. This distinction matters for audit, repair, and governance.

The mechanism addresses a gap that remains after metadata prompting, numerical provenance, and generic commit-time guards are applied separately. Metadata prompting can expose the information needed for a decision, but the model still determines probabilistically whether to use it. Numerical provenance records where a value came from, but does not establish that the current measurement semantics permit the requested transformation. Generic commit guards can ensure that relevant state is reconsidered

before an effect becomes durable, but they do not specify which statistical fields are relevant to a particular operation. SCAR-1.0 supplies that dependency specification, and the controller experiments measure how different choices of dependency scope change release behavior and revalidation work.

Design Implications

The results suggest that the model and the runtime should have different responsibilities in this workflow. The model can select an operation, compute the requested quantity, and provide an interpretation, but the decision to make that claim durable should depend on the state observed at commit. The available-channel sessions show that the models can reason productively over refreshed metadata. The controller is still needed because a staged decision should not remain sufficient for release after the evidence supporting that decision has changed.

The contract itself is part of the trusted state and therefore needs explicit version control. Each version should identify its required coordinates, decision order, canonicalization rules, and negative controls. Included coordinates should be backed by cases in which changing that coordinate changes authorization, while excluded coordinates should be tested with authority-preserving controls. A change to the contract must also invalidate receipts whose interpretation depends on the previous version, even if the underlying measurement record is unchanged. With these records preserved, a receipt identifies the contract version and measurement state under which a particular claim was authorized.

Operation-scoped dependencies also permit selective reuse of prior authorization results. If the receipt projection is unchanged, the metadata relevant to the requested operation is unchanged and the previous authorization can be reused. If the projection changes, the controller does not reject the claim automatically; it evaluates the claim again against the current state. This distinction is important because revalidation after a relevant change preserves valid releases that would otherwise be lost under simple invalidation, while still blocking release when a required premise has disappeared.

Keeping REJECT and ABSTAIN separate is also useful after a claim is blocked. REJECT identifies an observed contradiction and the field responsible for it, so resolution generally requires changing the task, data, or interpretation. ABSTAIN instead indicates that a required premise is unavailable, in which case retrieval or escalation may restore enough evidence to decide the claim. Collapsing the two states into a generic refusal would preserve the binary release boundary but remove information needed to choose the next action.

Limitations

First, SCAR-1.0 is author-defined and narrow. Standards motivate its coordinates and operator discipline, but no independent panel of statisticians validated the exact rules. Agreement between two deterministic implementations is not human semantic validation. Broader domains would require governed contracts and expert review.

Second, the stateful environment is controlled. It observes real model tool calls, actual version changes, and controller commits, but not uncontrolled drift in a provider's production data service or real deployment traffic. Values are intentionally fixed to isolate semantic authorization. Only 23 of 80 conflict values are observed elsewhere in source-backed records; the rest are schema-valid controlled combinations. The 18-set observed-value subset is supportive, not a substitute for field deployment.

Third, current-state access is bundled with a tool affordance whose name and description indicate that it reads state after a version change. Removing the user-message notice does not fully remove that information. The six controller policies are also deterministic branches applied to the same attempted effects, not independently randomized end-to-end deployments. Their safety, coverage, and evaluation counts describe SCAR-1.0 under the balanced transition design; they are not estimates of production frequencies, latency, or cost.

Fourth, four models appear in the static first-run comparison, but only Sol and Claude receive stochastic replication and live testing. Replications quantify output variability for those provider aliases on September 1, 2026; they do not identify future model-version variation. Provider family, training lineage, serving stack, and capability remain partly confounded.

Fifth, exact effect binding treats even a numerically close recomputation as a different staged effect. This is intentional for authorization integrity and exposes two valid-channel blocks, but applications may need a governed canonical numeric representation. The SHA-256 digest is only an experimental commitment; deployment also needs the controls identified in the trust model.

A de-identified reproducibility package containing SCAR-1.0, prompts, tool schemas, locked inputs, complete outputs, evaluation code, execution settings, request identifiers, and provenance manifests is available from the authors upon reasonable request.

Ethical Statement

The study uses public aggregate economic observations and no personal data. Its principal risk is overgeneralization. A REJECT decision means that a task violates SCAR-1.0, not that every statistical authority would reject the calculation. Every experimental record preserves the operation, contract

version, decisive field, state version, and interpretation boundary. We distinguish ABSTAIN from REJECT and report deterministic implementation agreement without inflating it into human validation or deployment assurance.

Conclusion

Correct numbers do not carry permanent permission. Across the matched live sessions, every invalid terminal action changes from RELEASE without current-state access to WITHHOLD when current state can be read. Because the numerical values are fixed across the transition, payload identity alone cannot expose the stale authorization. In the controller comparison, operation-scoped receipts block every invalid executable attempt observed in the benchmark while preserving every valid executable attempt and performing fewer contract evaluations than full-state or always-revalidate policies. The practical implication is that a durable statistical claim should remain bound not only to its numerical payload, but also to the current measurement state required by the operation that produced it.

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